PIYUSH YADAV

+91 8595955136 yadavpiyush1010@gmail.com LinkedIn Github

Summary

B.Tech Software Engineering graduate from Delhi Technological University with hands-on experience in Python, SQL, machine learning, and AI through projects in fraud detection, recommendation systems, and predictive analytics. Skilled in data preprocessing, EDA, feature engineering, model development, evaluation, and data visualization, with experience applying machine learning and Generative AI to data-driven problems.

Technical Skills

Programming and Querying: Python (Pandas, NumPy), SQL, MySQL, PostgreSQL, MongoDB
Machine Learning: Scikit-learn, XGBoost, Classification, Feature Engineering, Feature Selection, Model Evaluation, Hyperparameter Tuning
Data Science and Statistics: Data Preprocessing, Data Cleaning, Exploratory Data Analysis (EDA), Statistical Analysis, Data Transformation, Predictive Modeling
AI and Generative AI: Generative AI, Large Language Models (LLMs), Prompt Engineering
Data Visualization and Tools: Power BI, Tableau, Matplotlib, Seaborn, Jupyter Notebook, Git, GitHub, REST APIs

Projects

Banking Fraud Detection Analytics | *SQL · Power BI · Python · Scikit-learn · XGBoost* | GitHub

- Analyzed 300K+ banking transactions using SQL and Python, performing data preprocessing, exploratory data analysis (EDA), and feature engineering to identify fraudulent transaction patterns.
- Engineered 25+ transaction risk features and developed an XGBoost fraud detection model achieving 84% accuracy, 81% precision, and 87% recall for identifying fraudulent transactions.
- Evaluated fraud patterns through an interactive Power BI dashboard tracking Fraud Rate, Merchant Risk, Geographic Hotspots, and Transaction KPIs to support fraud monitoring and risk analysis.

E-commerce Product Recommendation System | *Python · SQL · Pandas · Scikit-learn · Power BI* | Project Link

- Analyzed 1M+ user interactions and product ratings using SQL and Python, performing data preprocessing and exploratory analysis to identify customer preferences and behavioral patterns.
- Developed and evaluated Collaborative Filtering, Content-Based, and Hybrid Recommendation Models, achieving 0.91 Precision through model evaluation and hyperparameter tuning.
- Built an interactive Power BI dashboard tracking Precision, CTR, user engagement, and top recommended products to evaluate recommendation performance and personalization.

Loan Approval System | *Python · SQL · Pandas · Scikit-learn · XGBoost · Power BI* | Project Link

- Analyzed 200K+ loan applications using SQL and Python, performing data preprocessing, feature engineering, and exploratory data analysis to identify factors influencing loan approval outcomes.
- Developed and compared Logistic Regression, Random Forest, and XGBoost models, achieving 87% accuracy and 0.89 ROC-AUC through hyperparameter tuning and model evaluation.
- Built an interactive Power BI dashboard tracking Approval Rate, Default Risk, Credit Score Distribution, and Applicant Segmentation to support data-driven lending analysis.

Certifications and Achievements

- IBM Data Science Professional Certificate — Python, SQL, machine learning, and data visualization
- Career Essentials in Generative AI by Microsoft and LinkedIn — Generative AI and AI fundamentals
- Google Prompt Engineering Certificate — Generative AI and prompt engineering
- NPTEL Certification in Python Programming — Python programming and data analysis
- Cisco Data Analytics Essentials — Data analysis and visualization
- Solved 250+ SQL and analytics problems across LeetCode and HackerRank.

Education

Delhi Technological University **New Delhi 2022–2026**
Bachelor of Technology (B.Tech), Software Engineering

Position of Responsibility

Cultural Council Society, DTU — Head
Led a team of 40–50 members and coordinated 11+ college initiatives and cultural events, contributing to large-scale events with 40K–50K+ total participants/attendees.